How useful are historical data for forecasting returns?

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Bocconi University: May 27, 2011

Focus of my recent research

Forecasting the dynamics of asset returns,
identifying, measuring and managing risk that results from those dynamics

- Recent research projects include:
  - Real-time identification of structural breaks in return distributions
  - Duration-dependent mixtures & Jump-GARCH mixtures
  - Forecasting time-varying risk premia
  - Pricing jump and crash risk
  - Stock market phases: bear market rallies and bull market corrections
  - Model risk: learning about model instability in real time

Some papers developing the approach

This presentation will draw from parts of the following two research papers:

- Forecasting long-run return distributions allowing for breaks
  - How useful are historical data for forecasting the long-run equity return distribution? (with Maheu; JBES, 2009)

- Do time-varying higher-order moments affect a distribution's location?
  - Do Jumps Contribute to the Dynamics of the Equity Premium? (with Maheu & Zhao, manuscript, 2010)

Some related papers in our research program on forecasting return distributions

- Components of market risk and return (with Maheu; JFEC, 2007)

- Components of bull and bear markets: bull corrections and bear rallies (with Maheu and Song, manuscript, 2010)

- Do high-frequency measures of volatility improve forecasts of return distributions? (with Maheu; JE, 2011)

Forecasting Distributions

We often require a forecast of the long-run (unconditional) return distribution

- For risk measurement, e.g. dispersion or tail area
- For long-run asset allocation or investment decisions e.g. pensions

Problem

- The shape of that distribution is sensitive to the estimation sample

Sensitivity of the distribution to the estimation sample

Table: Summary Statistics for Scaled Monthly Excess Returns

Sample	Mean	Variance	Skewness	Kurtosis
1885-2003	0.052	0.401	-0.452	9.950
1940-1959	0.118	0.249	-1.451	10.160
1960-1979	0.017	0.266	-0.264	3.878

How useful are historical data for forecasting the long-run equity return distribution?

John M. Maheu Thomas H. McCurdy

Bocconi University: May 27, 2011

Possibility of Structural Breaks

What might explain the above sensitivity to the sub-sample of data?

- Possibility of structural breaks in the distribution of market excess returns
- If this is the case, how should we adjust our estimator for density forecasts?
  - Which model or density function should we use?
  - How many historical data should we use to estimate model parameters?
  - How should we update forecasts as new data arrive?
- We need real-time out-of-sample density forecasts allowing for structural breaks

Then we can:

- Investigate implications for the long-run equity premium & risk measures

Some alternative approaches to structural breaks for stocks

Structural breaks in the equity premium:

- Pastor and Stambaugh (2001): economically motivated prior beliefs
- Kim, Morley, and Nelson (2005): links premium to shifts in variance
- Usual features:
  - In-sample (smoothed) estimates with structural breaks (not forecasts)
  - Chib (1998) model: estimates conditional on given number of breaks
  - Estimates do not integrate out uncertainty about number of breaks
  - No heteroskedasticity within regimes

Breaks in conditioning information:

- Lettau and van Nieuwerburgh (2006): mean of dividend-price ratio
- Paye and Timmermann (2006): multivariate conditional return model
- Viceira (1997): slope coefficient on dividend yield

Structural breaks in the market excess return distribution:

- Maheu and McCurdy (2009): density forecasts with learning about breaks

Other applications

Some other applications of structural breaks:

- Andreou and Ghysels (2002): breaks in volatility
- Lettau, Ludvigson and Wachter (2006): macroeconomic risks
- Maheu and Gordon (2007): application to macro-economic variables
- Pesaran and Timmermann (2007), (2002): optimal window length
- Pesaran, Pettenuzzo and Timmermann (2006b): pricing
- Pettenuzzo and Timmermann (2005): optimal asset allocation

Learning about breaks in real time

Proposal: Real-time out-of-sample density forecasts allowing for structural breaks

- Byproduct: identifies (and updates) probable break dates

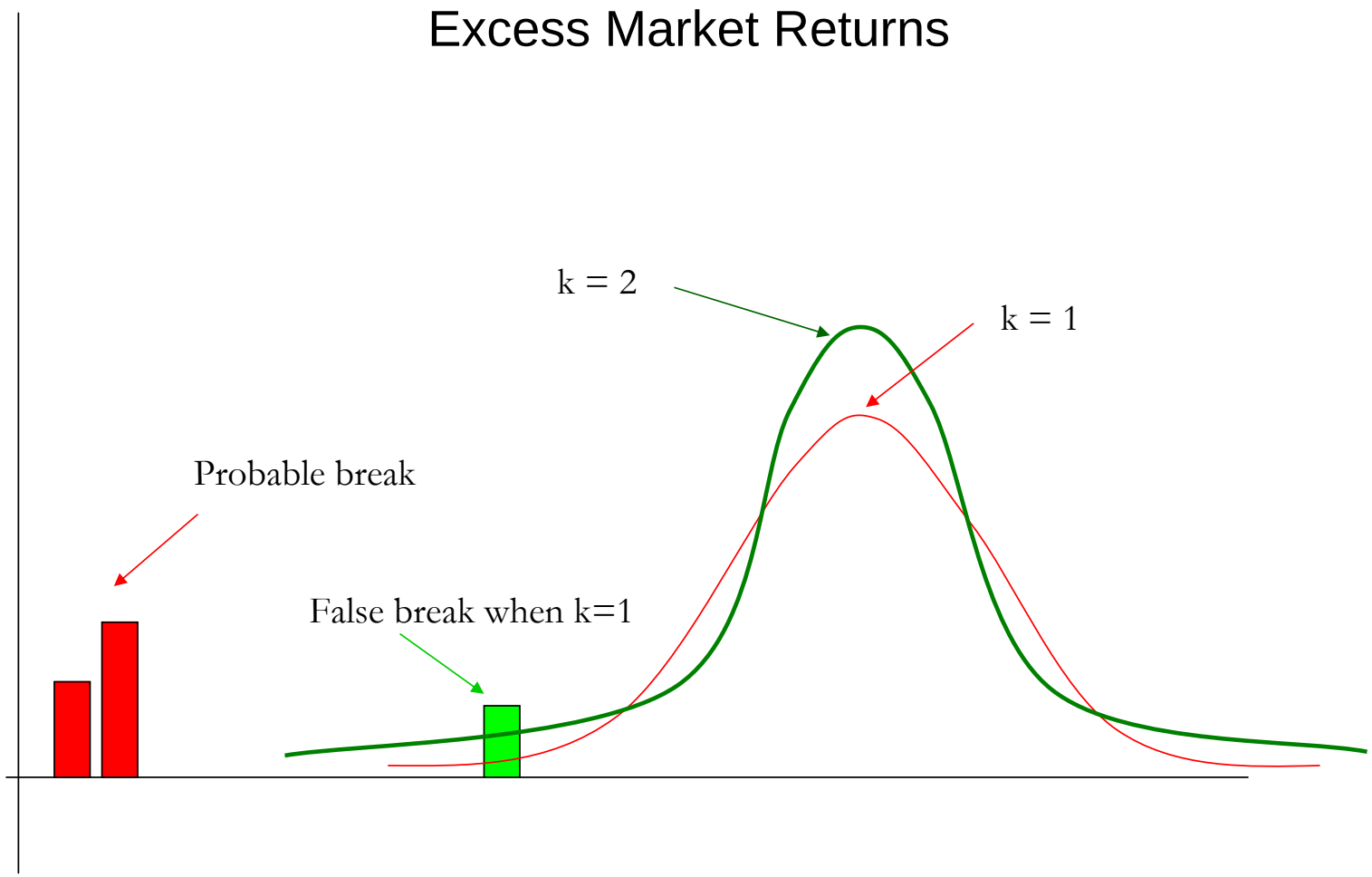
We postulate that the unconditional distribution of returns is generated by a

- Discrete mixture of normal distributions subject to occasional breaks
- Breaks governed by an i.i.d. Bernoulli distribution with parameter λ
- Implies time-varying and nonstationary unconditional distribution

How do we do inference in this situation?

- Breaks affect the parameter vector of the given submodel specification
- Use a Bayesian model of learning about structural breaks as new data arrive
  - View the data as being partitioned (by the breaks) into regimes
  - Stationary *submodels* describe data within regimes
- Estimate the probability of breaks in real time
- Probabilistic approach to both past and future breaks

Choice of Specification for the Submodel Structure



Submodel Specification

- k-component discrete mixture model of returns can be represented as

$$r_t = \begin{cases} N(\mu_1, \sigma_1^2) & \text{with probability } \pi_1 \\ \vdots & \vdots \\ N(\mu_k, \sigma_k^2) & \text{with probability } \pi_k, \end{cases}$$

with $\sum_{j=1}^{k} \pi_j = 1$.

- $k = 1$ is Gaussian distribution
- $k > 1$ can capture unknown forms of heteroskedasticity, skewness, fat tails

- The equity premium is: $\gamma = Er_t = \sum_{i=1}^{k} \mu_i \pi_i$
- Uncentered moments are $\gamma_j^{'} = Er_t^j$, $j = 2, 3, 4$; and
- Centered moments are $\gamma_2 = \gamma_2^{'} - (\gamma)^2$, etc.

- **For the rest of this discussion we will focus on the $k = 2$ case**

Individual Submodels and the Submodel Average

Submodel
– New submodels introduced periodically to check for breaks
– Submodels are indexed by introduction time i and parameter vector θ
– Submodel M_i allows for possibility that break occurs at time i

– Estimate θ using different submodels (histories of data) to allow for breaks
For example, at time T:

- Submodel M_1: estimate θ using $t = 1, ..., T$; compute predictions
- Submodel M_2: estimate θ using $t = 2, ..., T$; compute predictions
- etc
- Submodel M_T: estimate θ using $t = T$; compute predictions
- Submodel M_{T+1}: no data: just use prior to compute predictions

$\longrightarrow$ Submodel average combines all submodel predictions in an optimal sense, i.e. optimally selects submodels (data histories) based on their predictive content.

▸ Combining Submodels

Estimating the Probability of a Break

Arrival of breaks is i.i.d. Bernoulli with parameter λ

- Given a prior $p(\lambda)$, we can update beliefs given sample data
- The break process can be separated from the submodel estimation
  - The posterior of submodel parameters is independent of λ
  - Posterior for λ is a function of submodel predictive likelihoods which have parameter uncertainty integrated out
- Therefore likelihood is function of only 1 parameter so the posterior is

$$p(\lambda|I_{t-1}) \propto p(\lambda) \prod_{j=1}^{t-1} p(r_j|I_{t-j}, \lambda)$$

Forecasting in the presence of learning about breaks

Our forecasts use a probability-weighted average of submodel forecasts

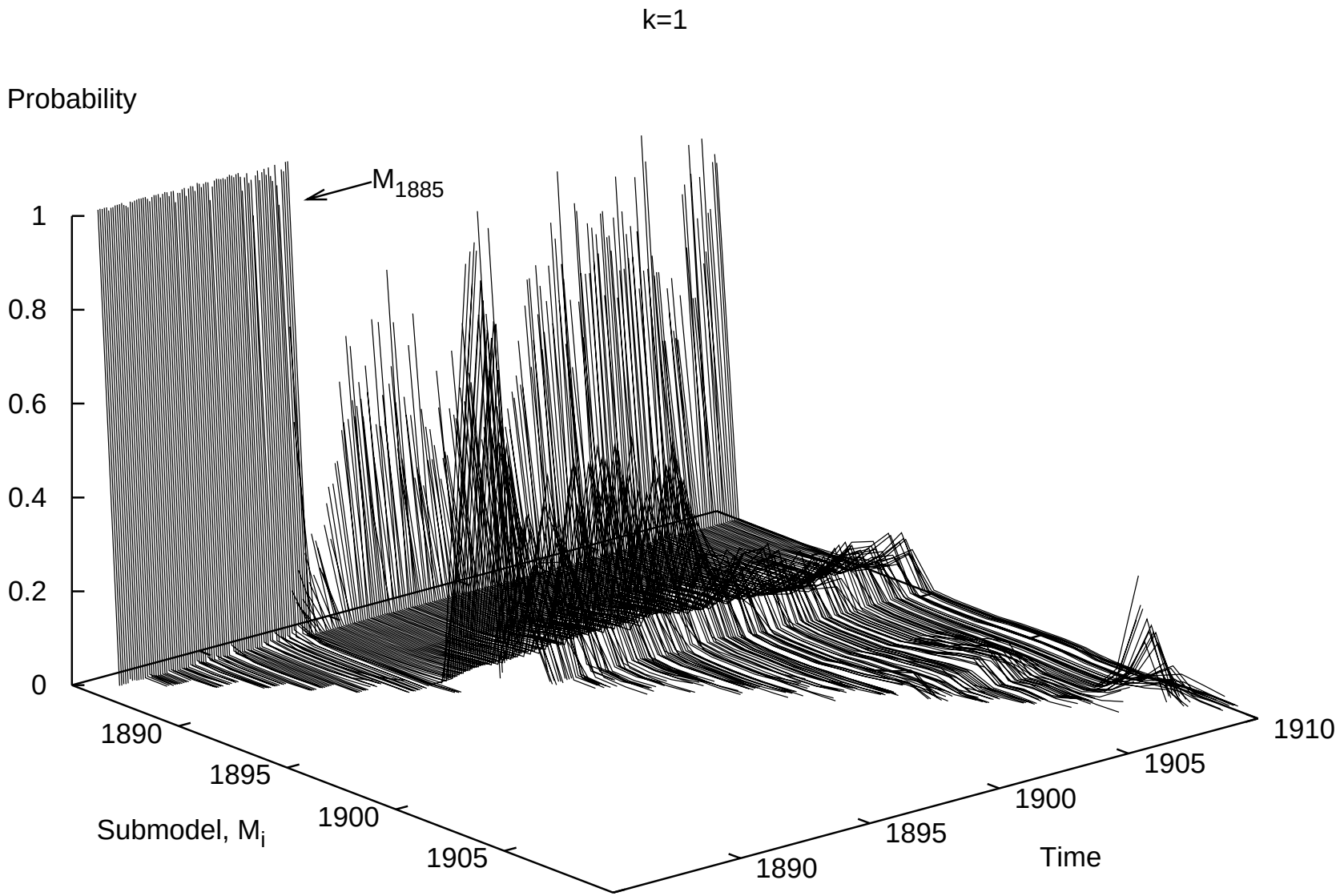
Our structural break model:

- Estimates submodel probabilities at each t based on predictive content
- Use submodel probabilities and λ_t to combine the submodel forecasts
- Optimal use of historical data (we show that fixed window is not optimal)

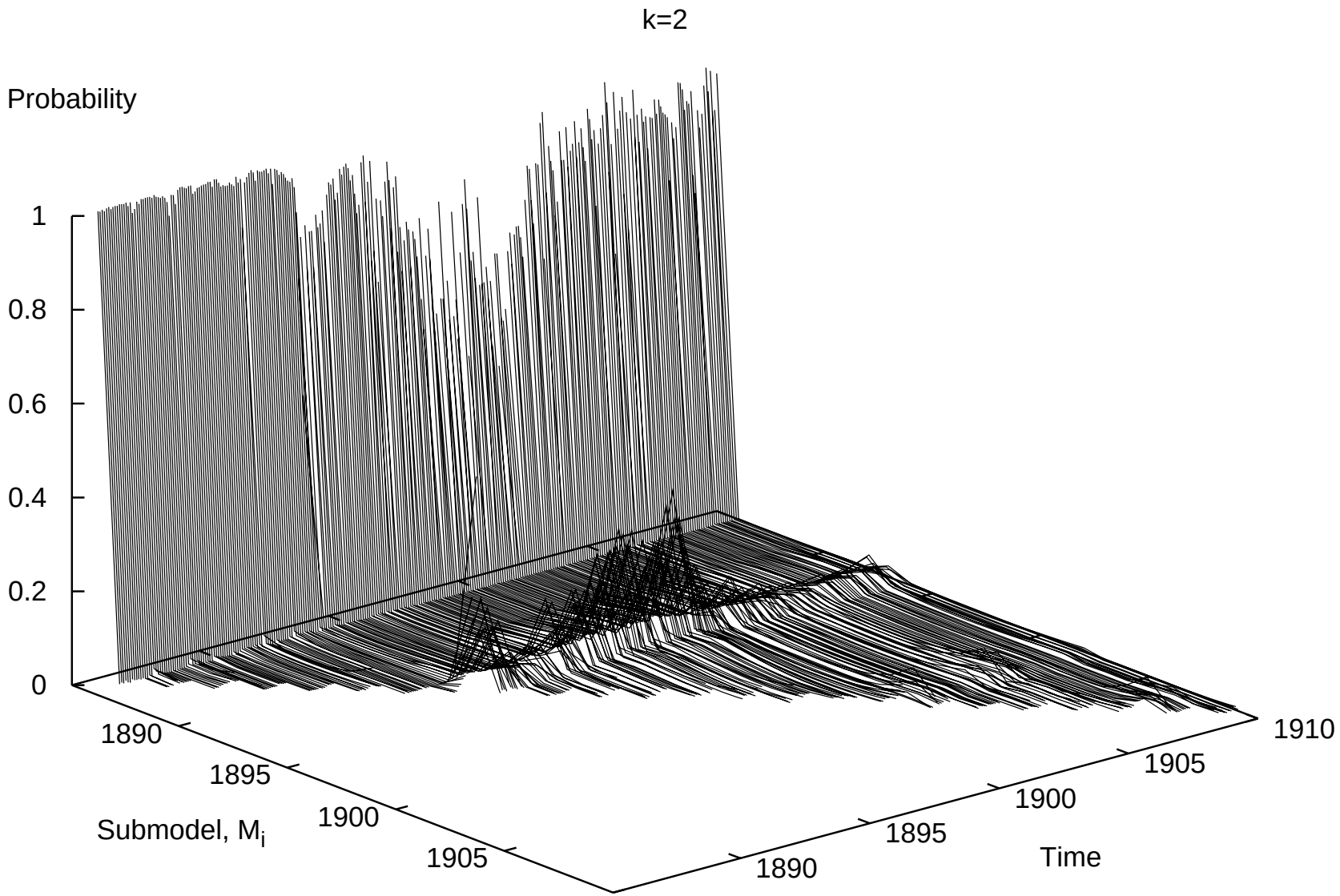
- Our forecasts incorporate time-varying submodel uncertainty
  - Selects submodels (data histories) based on predictive content
  - We learn about breaks over time (time-varying submodel uncertainty)
  - Submodel probability distribution provides estimate of most recent break dates
  - Submodel averaging integrates out break points

▸ Predictive Likelihood

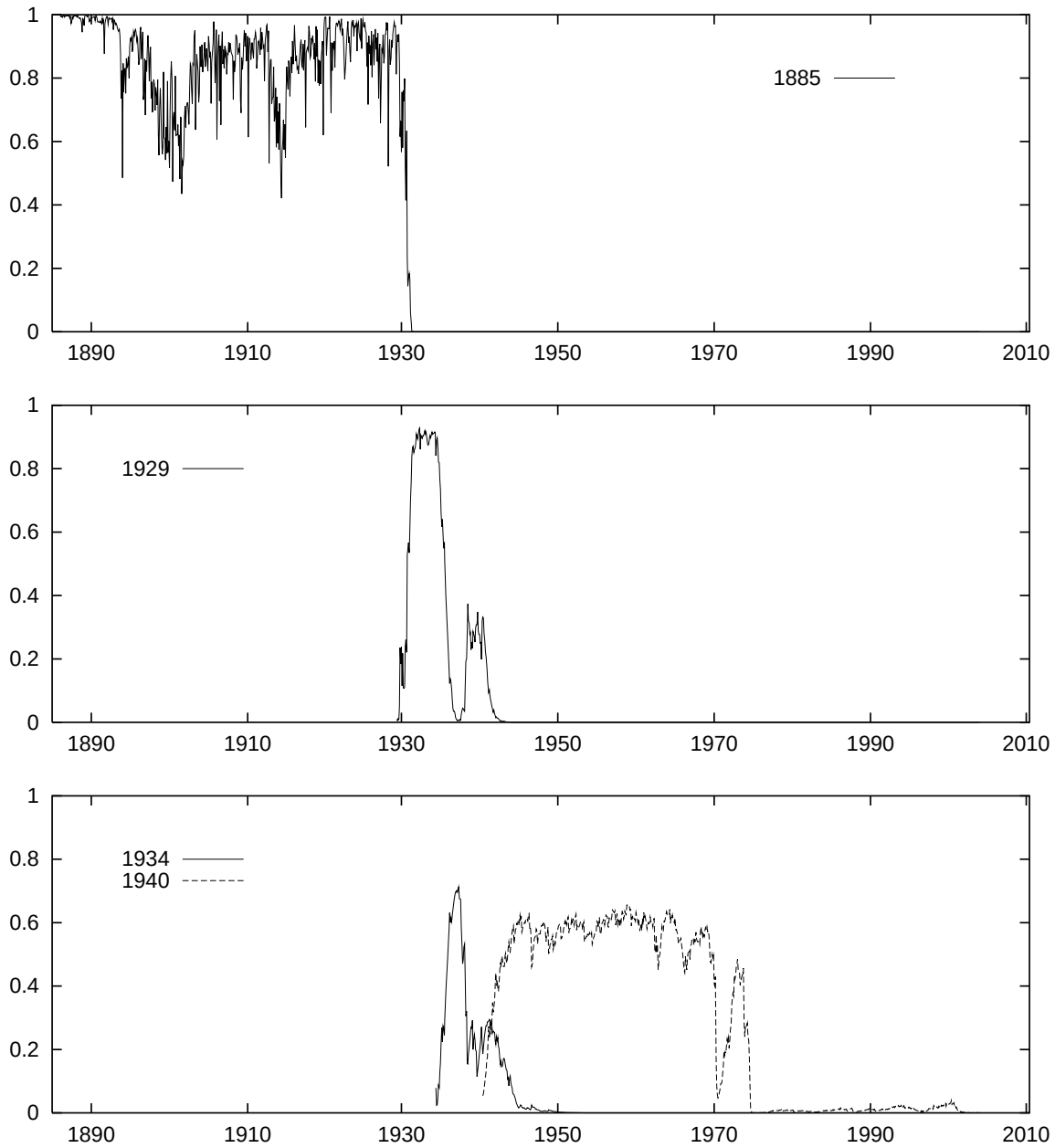
Submodel Probabilities, 1885-1910, ($k = 1$)



Submodel Probabilities, 1885-1910, ($k = 2$)



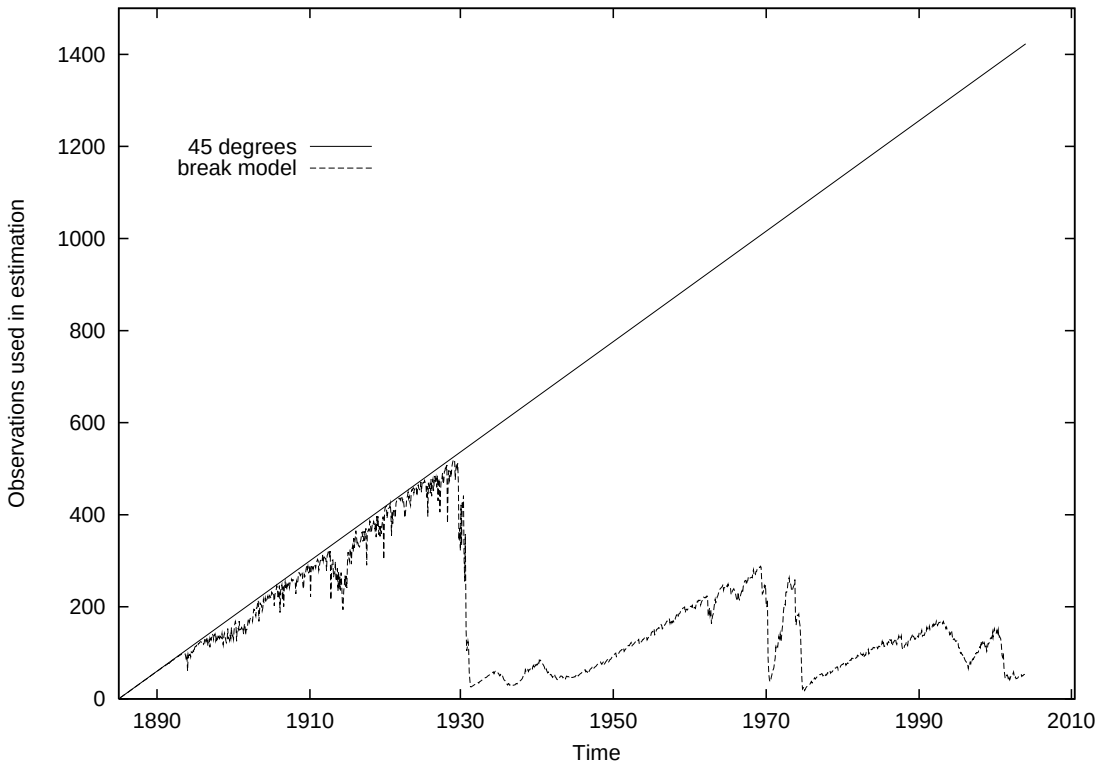
Example: Submodel Probabilities over Time ($k = 2$)



Results and Mean-Useful Observations, $k = 2$

Structural break model dominates using all data weighted equally or a fixed-window width

- $p(M_i|I_t)$ is the posterior submodel probability for M_i at time t
- $\sum_{i=1885}^{t} ip(M_i|I_t)$ is the mean of the submodel distribution at time t
- $\text{MUO}_t = \sum_{i=1885}^{t}(t+1-i)p(M_i|I_t)$,
  the expected number of useful observation for estimation at each t



Do jumps contribute to the dynamics of the equity premium?

John M. Maheu Thomas H. McCurdy Xiaofei Zhao

Bocconi University: May 27, 2011

Motivation: Using historical data optimally

Theme: Don't treat individual moments as separate 'parameters'; focus on the entire distribution

In the previous paper

- We focused on potential breaks as source of changing shape of the distribution of returns
- Real-time density forecasts from our break model were superior to alternatives

In this paper

- We focus on jumps as a source of changes in the shape of the distribution
- Do changes in the shape of the distribution affect dynamics of the equity premium?
- Interaction between higher-order moments and changes in location of the distribution

Why focus on jumps in this paper?

Jumps are large discrete changes in price

Jumps contribute to the conditional variance and also drive dynamics in the tail

Jumps matter for:

- Risk management: conditional VaR or expected shortfall, crash risk, etc.
- Intertemporal portfolio allocation
- Pricing
  - Help explain option pricing misspecification
  - **Do jumps contribute to risk premia for the underlying (equity returns)?**

How Do Jumps Affect Returns?

Jumps are one component of the conditional distribution of returns:

- Jumps impact the innovations to returns
- Jumps contribute to the level and dynamics of volatility
- Jumps are realizations in the tail of the distribution
  - Persistence associated with jump arrivals implies time-varying conditional skewness and time-varying conditional kurtosis
- Jumps may affect the location of the distribution
  - Are jumps priced through higher-order moments?

All effects important for forecasts and thus for financial management decisions

Example papers: jumps associated with equity returns

Time-varying jump arrival, for example:

- Continuous-time SV-jump diffusion specifications:

  Andersen, Benzoni and Lund (1999), Yu (1999), Bates (2000), Das (2002), Chernov, Gallant, Ghysels and Tauchen (2003), Eraker, Johannes and Polson (2003)

- Discrete-time models:

  Pan (1997), Bekaert and Gray (1998), Bates and Craine (1999), Johannes, Kumar and Polson (1999), Neely (1999), Chan and Maheu (2002), Maheu and McCurdy (2004)

Pricing jump risk using options and/or equity data:

- Pan (2002)
- Christoffersen, Jacobs and Ornthanalai (2008)
- Guidolin and Timmermann (2008)
- Elkamhi and Ornthanalai (2009)
- Duan and Zhang (2010)
- This paper (2010)

Definition of Excess Returns

Define the continuously compounded excess return on the market index as

$$r_t \equiv r_{m,t} - r_{f,t}$$

$r_{m,t} \equiv$ the continuously compounded market index return
$r_{f,t} \equiv$ the continuously compounded riskfree rate

Henceforth, we refer to r_t as the log return

Dynamics of Returns

Assume that the dynamics of realized log returns are driven by

$$r_t = m_t + u_t$$

$m_t \equiv$ expected market equity premium using info at time $t-1$

$u_t \equiv \rho_1 u_{t-1} + \rho_2 u_{t-2} + \epsilon_t$

$\epsilon_t \equiv$ mean zero return innovation

That is, excess returns are driven by:

- The expected market equity premium m_t
- An autoregressive component due to stale prices or missing pricing factors
  Note: we test for robustness to including or excluding this AR structure
- And a mean zero return innovation ϵ_t

Decomposition of the return innovation

As in Maheu and McCurdy (2004), return innovations are hypothesized to be driven by:

- Normal stochastic changes
- Jumps (time-varying arrival of jump events plus stochastic jump size)

Both components are directed by the dynamics of the latent information process
(news arrival and dissemination)

▸ Examples of news associated with jump size and clustering

Total return innovations:

$$\epsilon_t \equiv \epsilon_{1,t} + \epsilon_{2,t}$$

Normal innovations:

$$\epsilon_{1,t} = \sigma_t z_t, \quad z_t \sim NID(0,1), \quad \epsilon_{1,t}|\Phi_{t-1} \sim N(0, \sigma_t^2)$$
$$E\left[\epsilon_{1,t}|\Phi_{t-1}\right] = 0$$

Abnormal or (compensated) jump innovations:

$$\epsilon_{2,t} = J_t - E\left[J_t|\Phi_{t-1}\right]$$
$$E\left[\epsilon_{2,t}|\Phi_{t-1}\right] = 0$$

Parameterization of the jump frequency distribution

- Define the discrete-valued number of jumps over $(t-1,t)$ as $n_t \in 0,1,2,...$
- The conditional distribution of n_t is Poisson with parameter λ_t

$$P(n_t = j|\Phi_{t-1}) = \frac{\exp(-\lambda_t)\lambda_t^j}{j!}, j = 0,1,2,....$$

$$E[n_t|\Phi_{t-1}] = \lambda_t$$

$n_t \equiv$ the number of jumps in period t

$\lambda_t \equiv$ the expected number of jumps for period t given time $t-1$ information

Time-series parameterization of the jump arrival process

- We parameterize the time-series dynamics of λ_t as an autoregressive process (Chan and Maheu (2002) & Maheu and McCurdy (2004))

$$\lambda_t = \gamma_0 + \gamma_1 \lambda_{t-1} + \gamma_2 \zeta_{t-1}$$

- Jumps are latent
  expected number of jumps is computed using the estimation filter

- The jump arrival *innovation* for period $t-1$ is defined as

$$\begin{aligned} \zeta_{t-1} &= E[n_{t-1}|\Phi_{t-1}] - \lambda_{t-1} \\ &= E[n_{t-1}|\Phi_{t-1}] - E[n_{t-1}|\Phi_{t-2}] \end{aligned}$$

- That is, the update in the expected number of jumps for period $t-1$ when information is updated from $t-2$ to $t-1$

- Note that jump arrivals are directed by the jump arrival innovations not by the return innovations that direct the GARCH process

Parameterization of the jump-size distribution

- $Y_{t,k} \equiv$ the size of jump k in period t drawn from $Y_{t,k} \sim N(\theta, \delta^2)$
- $J_t \equiv$ the total size of jumps in period t

$$J_t = \sum_{k=1}^{n_t} Y_{t,k}$$

- $\epsilon_{2,t} \equiv$ the compensated jump innovation to returns

$$\epsilon_{2,t} = J_t - \theta\lambda_t$$

- Since $E[J_t|\Phi_{t-1}] = \theta\lambda_t$, the compensated jump innovation is mean zero

$$E[\epsilon_{2,t}|\Phi_{t-1}] = E[J_t|\Phi_{t-1}] - \theta\lambda_t = 0$$

The volatility clustering directing process

- The normal innovation to returns, $\epsilon_{1,t}$, is assumed to be directed by a 2-component GARCH specification with feedback from jumps

$$\epsilon_{1,t} = \sigma_t z_t, \quad z_t \sim NID(0,1), \quad \epsilon_{1,t}|\Phi_{t-1} \sim N(0, \sigma_t^2)$$

- σ_t^2 is directed by a two-component GARCH process specified as:

$$\begin{aligned}
\sigma_t^2 &= \sigma_{1,t}^2 + \sigma_{2,t}^2 \\
\sigma_{1,t}^2 &= \omega + g_1(A_1, \Phi_{t-1})\epsilon_{t-1}^2 + \beta_1 \sigma_{1,t-1}^2 \\
\sigma_{2,t}^2 &= g_2(A_2, \Phi_{t-1})\epsilon_{t-1}^2 + \beta_2 \sigma_{2,t-1}^2 \\
\epsilon_{t-1} &= \epsilon_{1,t-1} + \epsilon_{2,t-1}
\end{aligned}$$

- Other 2-component volatility clustering models:
  Engel and Li (1999), Maheu and McCurdy (2007), Chan and Feng (2009)

News impact parameterization

The generalized news impact coefficient for the ith GARCH component is

$$\begin{aligned} g_i(A_i, \Phi_{t-1}) &= \exp(\alpha_i + I(\epsilon_{t-1})(\alpha_{a,j,i} E[n_{t-1}|\Phi_{t-1}] + \alpha_{a,i})), i = 1, 2 \\ I(\epsilon_{t-1}) &= 1 \text{ if } \epsilon_{t-1} < 0, \text{otherwise} \quad I(\epsilon_{t-1}) = 0 \end{aligned}$$

Note that this parameterization incorporates:

- Asymmetric impact from good versus bad news
- Asymmetric feedback to future vol from jumps vs normal innovations (conditional on bad news)
  - Note that $E[n_{t-1}|\Phi_{t-1}]$ is the estimated number of jumps at time $t-1$

Dynamics of Higher-Order Conditional Moments

- Analogous to derivation in Das (2002) & Maheu and McCurdy (2004):

$$
\begin{aligned}
v_t &= \sigma_t^2 + \lambda_t(\theta^2 + \delta^2) \equiv \text{conditional variance} \\
s_t &= \frac{\lambda_t(\theta^3 + 3\theta\delta^2)}{(\sigma_t^2 + \lambda_t\theta^2 + \lambda_t\delta^2)^{3/2}} \equiv \text{conditional skewness} \\
k_t &= 3 + \frac{\lambda_t(\theta^4 + 6\theta^2\delta^2 + 3\delta^4)}{(\sigma_t^2 + \lambda_t\delta^2 + \lambda_t\theta^2)^2} \equiv \text{conditional kurtosis}
\end{aligned}
$$

- If $\lambda_t \equiv \lambda > 0$, that is, constant jump arrival rate
  still have time-varying and non-normal levels of skewness and kurtosis
- If $\lambda_t = 0$ for all t, that is, no jumps,
  conditional skewness and conditional kurtosis would be the same
  as that for a conditional normal distribution

Specification for the equity premium

What functional form should one use for the equity premium?

▸ Harvey and Siddique, 2000

- A Taylor-series approximation to a nonlinear pricing kernel implies a 4-moment parameterization (Guidolin and Timmermann, 2008)
- Restriction implied by decreasing absolute prudence $\psi_s \leq 0, \psi_k \geq 0$ (Dittmar, 2002)

Our maintained parameterization for the market equity premium is:

- $m_t \equiv$ conditional market equity premium: estimate with and without an intercept μ

$$m_t = \psi_v v_t + \psi_s s_t + \psi_k k_t; \quad \psi_s \leq 0, \quad \psi_k \geq 0$$

Alternative parameterizations of the equity premium:

- The special case of power utility (Duan and Zhang, 2010)
  $m_t = \psi_v v_t + \psi_s v_t^{\frac{3}{2}} s_t + \psi_k v_t^2 k_t$, in which
  $\psi_v = (\gamma - 0.5)$, $\psi_s = -(3\gamma^2 - 3\gamma + 1)/6$, $\psi_k = (4\gamma^3 - 6\gamma^2 + 4\gamma - 1)/24$
- Finally, an unrestricted parameterization: $m_t = \psi_v v_t + \psi_s s_t + \psi_k k_t$

Maximum Likelihood Estimation

Given the number of jumps $j \in \{0, 1, 2, \dots\}$ and parameter vector Θ

the conditional density of returns $f(r_t|\Phi_{t-1}, \Theta, n_t = j)$ is Normal:

$$\frac{1}{\sqrt{2\pi(\sigma_t^2 + j\delta^2)}} \exp\left(-\frac{1}{2}\frac{(r_t - m_t - \rho_1(r_{t-1} - m_{t-1}) - \rho_2(r_{t-2} - m_{t-2}) - (j - \lambda_t)\theta)^2}{\sigma_t^2 + j\delta^2}\right)$$

The full likelihood contribution in terms of r_t is then

$$f(r_t|\Phi_{t-1}, \Theta) = \sum_{j=0}^{\infty} f(r_t|\Phi_{t-1}, \Theta, n_t = j) P(n_t = j|\Phi_{t-1}, \Theta)$$

in which $P(n_t = j|\Phi_{t-1}, \Theta)$ is the p.d.f. of the time-varying Poisson distribution

The full-sample loglikelihood $l = \sum_{t=1}^{T} \log f(r_t|\Phi_{t-1}, \Theta)$ is maximized wrt Θ

The filter is:

$$P(n_t = j|\Phi_t, \Theta) = \frac{f(r_t|n_t = j, \Phi_{t-1}, \Theta) P(n_t = j|\Phi_{t-1}, \Theta)}{f(r_t|\Phi_{t-1}, \Theta)}, j = 0, 1, 2, \cdots$$

▸ interpretation of mixture

Equity Premium: Alternative Specifications

- Dynamics of variance and skewness both important

$$
\begin{aligned}
r_t &= m_t + \rho_1(r_{t-1} - m_{t-1}) + \rho_2(r_{t-2} - m_{t-2}) + \epsilon_{1,t} + \epsilon_{2,t} \\
m_t &= \mu + \psi_v v_t + \psi_s s_t + \psi_k k_t
\end{aligned}
$$

	Pricing Models					
Parameter	prudence	intercept	unrestricted	variance	skewness	kurtosis
μ		-0.026	-0.015	0.022	-0.027	-0.013
		(-1.377)	(-0.866)	(3.562)	(-1.386)	(-0.691)
ρ_1	0.157	0.157	0.154	0.158	0.157	0.158
	(21.909)	(21.879)	(21.382)	(22.054)	(21.878)	(22.030)
ρ_2	-0.048	-0.049	-0.051	-0.047	-0.049	-0.048
	(-6.958)	(-7.000)	(-7.325)	(-6.788)	(-7.003)	(-6.872)
ψ_v	0.027	0.040	0.051	0.022	0.040	0.031
	(3.270)	(3.266)	(4.453)	(1.991)	(3.274)	(2.757)
ψ_s	-0.028	-0.054	-0.256		-0.055	
	(-3.438)	(-2.395)	(-3.337)		(-2.402)	
ψ_k	0.000	0.000	-0.021			0.004
	(0.000)	(0.000)	(-2.777)			(1.931)
lgl	-24577.290	-24576.271	-24571.976	-24580.631	-24576.268	-24578.235

Importance of Higher Order Moments in Equity Premium

Likelihood Ratio Test Statistics: LRT (p-value)

- Higher-order moments are significant drivers of the equity premium
  - No pricing of higher-order ($\psi_v = \psi_s = \psi_k = 0$): LRT=12.62 (0.0056)
  - No pricing of skewness or kurtosis ($\psi_s = \psi_k = 0$): LRT=8.720 (0.0128)

- At the market level skewness and kurtosis pick up the same source of risk
  - Kurtosis not priced ($\psi_k \approx 0$) in the presence of skewness in prudence model
  - Evident from high negative correlation (-90%) between skewness and kurtosis
  - Chang, Christoffersen and Jacobs (2009): high negative correlation between risk neutral skewness and kurtosis, kurtosis not robustly priced

- AR(2) structure in conditional mean important
  - No AR(2) structure ($\rho_1 = \rho_2 = 0$): LRT=498.26 (6.4e-109)

- Importance of higher-order moments is robust to inclusion of AR(2)

▸ No AR(2) and other equity premium specifications

▸ Variance Dynamics

Jump Dynamics and Jump Size

- Jumps highly persistent ($\hat{\gamma}_1$ and $\hat{\gamma}_2 \neq 0$)
- On average jumps capture the left tail ($\hat{\theta} < 0$)

$$\begin{aligned} \lambda_t &= E[n_t|\Phi_{t-1}], \quad \lambda_t = \gamma_0 + \gamma_1 \lambda_{t-1} + \gamma_2 \zeta_{t-1}, \\ \epsilon_{2,t} &= \sum_{k=1}^{n_t} Y_{t,k} - \theta\lambda_t, \quad Y_{t,k} \sim N(\theta, \delta^2) \end{aligned}$$

	Pricing Models					
Parameter	prudence	intercept	unrestricted	variance	skewness	kurtosis
γ_0	0.007	0.006	0.003	0.009	0.006	0.008
	(3.458)	(2.590)	(2.897)	(4.103)	(2.578)	(3.524)
γ_1	0.949	0.957	0.970	0.941	0.957	0.947
	(89.851)	(80.449)	(122.384)	(88.914)	(80.488)	(87.323)
γ_2	0.182	0.152	0.091	0.212	0.151	0.190
	(4.392)	(3.567)	(3.831)	(4.887)	(3.555)	(4.337)
θ	-0.467	-0.460	-0.446	-0.490	-0.460	-0.476
	(-9.529)	(-9.107)	(-8.148)	(-10.050)	(-9.095)	(-9.719)
δ	0.942	0.976	1.040	0.920	0.978	0.940
	(16.540)	(14.028)	(14.747)	(17.584)	(13.992)	(16.291)

Importance of Jump Dynamics and Variance Dynamics

Likelihood Ratio Test Statistics: LRT (p-value)

Jump Dynamics

- Jumps are important
  - No jumps ($\lambda_t = 0$): LRT=1459 (1.2e-308)
- Significant persistence in jump arrival
  - No predictable component ($\gamma_1 = \gamma_2 = 0$): LRT=103 (3.74e-23)
- Asymmetry associated with jumps is significant
  - No asymmetry ($\alpha_{a,j,1} = \alpha_{a,j,2} = 0$): LRT=37 (9.21e-09)

Variance Dynamics

- $2nd$ GARCH component is important
  - No $2nd$ GARCH component ($g_2(\Lambda, \Phi_{t-1}) = \beta_2 = 0$): LRT=246 (4.13e-52)
- Jumps are important to variance dynamics
  - No jumps and no $2nd$ GARCH component ($\lambda_t = g_2(\Lambda, \Phi_{t-1}) = \beta_2 = 0$): LRT=1872 (1.6e-394)

Summary Statistics for the Equity Premium

Summary Statistics: Daily Equity Premium m_t in Percentage

	Median	Mean	StDev
m_{prudence}	0.031	0.039	0.036
$m_{\text{unrestricted}}$	0.028	0.038	0.057

Summary Statistics: Higher-Order Moments

	Median	Mean	StDev
'Prudence model'			
Variance	0.518	0.947	1.465
Skewness	-0.451	-0.496	0.320
Kurtosis	4.669	5.238	2.117
'Unrestricted model'			
Variance	0.512	0.954	1.512
Skewness	-0.422	-0.466	0.311
Kurtosis	4.825	5.447	2.324

How Much Do Jumps Contribute to Equity Premium?

- Equity premium increases when the jump intensity increases
  - This effect is strongest when σ is small and/or jump intensity is small plot
- On average one more jump per year increases the equity premium by 10.57 bps
- The average contribution of jumps to the equity premium can be estimated as

$$\overline{\left(\frac{\partial m}{\partial \lambda}\right)} \times \bar{\lambda}$$

- Our estimate of this jump risk premium is 3.96% per annum on average
  - Other studies (using stocks and options) find similar results
    Pan (2002): 3.5%.
    Elkamhi and Ornthanalai (2009): 3.18%
- Approximately one-third (1.12% per annum) of the jump risk premium comes through contribution of jumps to variance dynamics
  - $\psi_v \times (\bar{\lambda} \times (\theta^2 + \delta^2)) \times 252$
- The remainder (2.84% per annum) of the jump risk premium comes through contribution of jumps to skewness

Skewness Premium

- In our parameterization, the skewness premium (annualized %) is:

$$\psi_s \times E[s_t] \equiv \psi_s \times \bar{s} \times 252$$

- Our estimate of this skewness risk premium is 3.5% on average

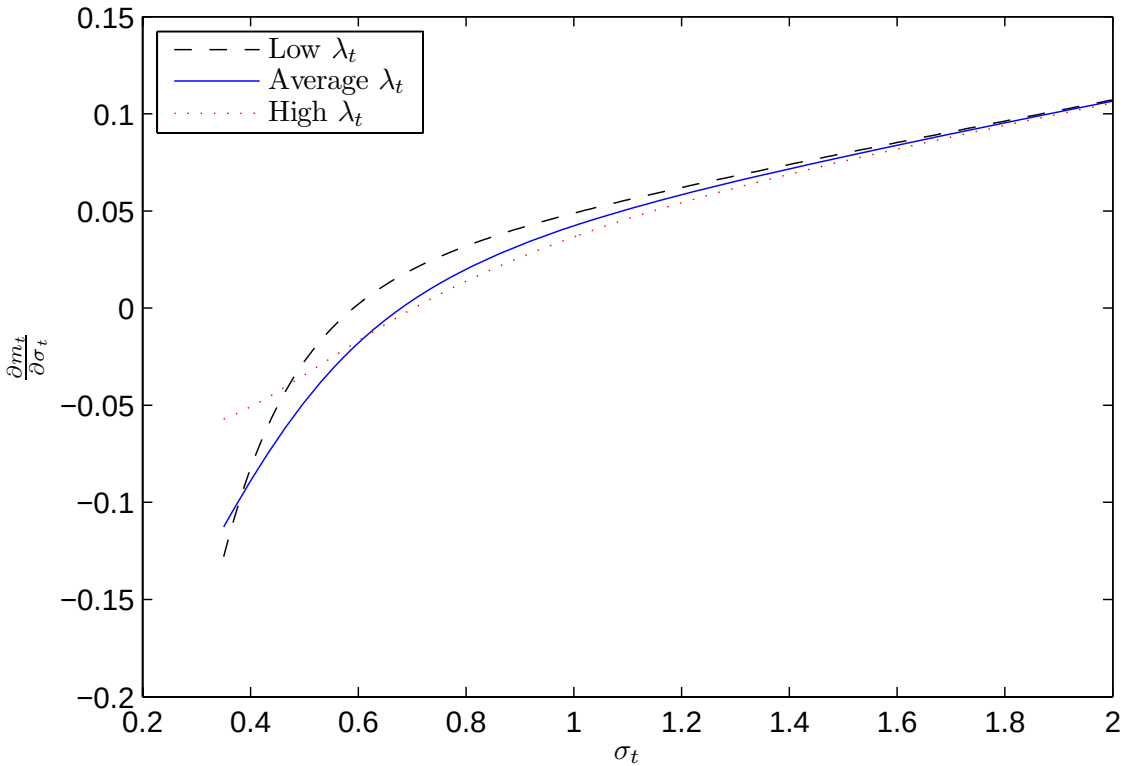
- Harvey and Siddique (2000) find a skewness premium of 3.6% in a cross-sectional study using monthly returns.

Traditional Market Risk-Return Tradeoff Revisited

Traditionally measured by sensitivity of the equity premium to variance clustering e.g. GARCH

In our case: $\partial m/\partial \sigma$ implies that:

- Increase in σ increases variance and decreases *negative* skewness (opposite effects on m)
- $\partial m/\partial \sigma < 0$ in calm times (small σ) and positive in more volatile times
- Changes from negative to positive at around sample mean of σ



Traditional Market Risk-Return Tradeoff Revisited, continued

In our case:

$$\begin{aligned} \partial m/\partial\sigma &= \partial\left(\hat{\psi}_v \times \hat{v} + \hat{\psi}_s \times \hat{s} + \hat{\psi}_k \times \hat{k}\right)/\partial\sigma \\ &\approx \partial\left(\hat{\psi}_v \times \hat{v} + \hat{\psi}_s \times \hat{s}\right)/\partial\sigma \\ &= \underbrace{\hat{\psi}_v}_{>0} \times \underbrace{\partial\hat{v}/\partial\sigma}_{>0} + \underbrace{\hat{\psi}_s}_{<0} \times \underbrace{\partial\hat{s}/\partial\sigma}_{>0} \end{aligned}$$

Note that $\hat{s} < 0$ since $\hat{\theta} < 0$; so $\partial\hat{s}/\partial\sigma > 0$ refers to a decrease in negative skewness

But, traditionally, only σ is included when testing the market risk-return tradeoff

- During calm times (low level of σ_t):
  - Although $\partial v/\partial\sigma > 0$, the skewness premium dominates so $\partial m/\partial\sigma < 0$
  - Therefore, if s_t is omitted from the pricing relationship,
    the relationship between variance and return can be forced to be negative
- During volatile times: $\partial m/\partial\sigma > 0$
  - Regardless of whether or not s_t is included
    will find a positive relationship between variance and return

Concluding Comments

We show that higher-order moments are priced in the market return

- Time varying higher-order moments are driven by
  AR jump arrival, stochastic jump size, and 2-component GARCH dynamics
- Jumps increase the equity premium (jump risk premium of 3.96% per annum)
  - Approximately one-third (1.12% per annum) of the jump risk premium comes through contribution of jumps to variance dynamics
  - The remainder (2.84% per annum) of the jump risk premium comes through contribution of jumps to skewness
- Skewness and kurtosis pick up the same crash risk at the market level
- Skewness premium of 3.5% per annum
- Pricing higher order moments offers a potential resolution to conflicting results on risk-return relationship
- Possible to identify jump risk's contribution to the equity premium using equity data only

Submodels: Priors, Posteriors and Predictive Densities

- Conditional data density for submodel M_i given θ and $I_{i,t-1}$ is $p(r_t|\theta, I_{i,t-1}, M_i)$
- Prior distribution for the parameter vector θ of submodel M_i is $p(\theta|M_i)$
- Posterior density for each possible submodel is

$$p(\theta|I_{i,t-1}, M_i) \propto \begin{cases} p(r_i, ..., r_{t-1}|\theta, M_i)p(\theta|M_i) & i < t \\ p(\theta|M_i) & i = t, \end{cases}$$

  $i = 1, ..., t.$

- Each M_i has a specific predictive density for r_t

$$p(r_t|I_{i,t-1}, M_i) = \int p(r_t|I_{i,t-1}, \theta, M_i)p(\theta|I_{i,t-1}, M_i)d\theta, \quad i = 1, ..., t.$$

Combining Submodels

Given λ_t, $0 \leq \lambda_t \leq 1$, $t = 1, ..., T$, consider forecasts for $t = 2, 3, ...$

- At $t = 1$:

$$\begin{aligned} p(r_2|I_1, \Lambda_2) &= p(r_2|I_{1,1}, M_1)p(M_1|I_1, \Lambda_1)(1-\lambda_2) + p(r_2|I_{2,1}, M_2)\lambda_2 \\ &\quad \text{(P.D. given no break at t=2)} \times \text{(prob no break at t=2)} \\ &+ \quad \text{(P.D. given break at t=2)} \times \text{(prob break at t=2)} \end{aligned}$$

- Given r_2, we can update submodel probabilities:

$$p(M_1|I_2, \Lambda_2) = \frac{p(r_2|I_{1,1}, M_1)p(M_1|I_1, \Lambda_1)(1-\lambda_2)}{p(r_2|I_1, \Lambda_2)}$$

$$p(M_2|I_2, \Lambda_2) = \frac{p(r_2|I_{2,1}, M_2)\lambda_2}{p(r_2|I_1, \Lambda_2)}$$

- Note that $I_{2,1} = \{\emptyset\}$ and $\Lambda_t = \{\lambda_2, ..., \lambda_t\}$

Combining Submodels, continued

At $t = 2$:

$$\begin{aligned} p(r_3|I_2, \Lambda_3) &= [p(r_3|I_{1,2}, M_1)p(M_1|I_2, \Lambda_2) + p(r_3|I_{2,2}, M_2)p(M_2|I_2, \Lambda_2)]\,(1 - \lambda_3) \\ &+ p(r_3|I_{3,2}, M_3)\lambda_3; \qquad I_{3,2} = \{\emptyset\} \end{aligned}$$

(P.D. given no break at t=3) $\times$ (prob no break at t=3)

(P.D. given break at t=3) $\times$ (prob break at t=3)

- After observing r_3, the updated submodel probabilities are

$$\begin{aligned} p(M_1|I_3, \Lambda_3) &= \frac{p(r_3|I_{1,2}, M_1)p(M_1|I_2, \Lambda_2)(1 - \lambda_3)}{p(r_3|I_2, \Lambda_3)} \\ p(M_2|I_3, \Lambda_3) &= \frac{p(r_3|I_{2,2}, M_2)p(M_2|I_2, \Lambda_2)(1 - \lambda_3)}{p(r_3|I_2, \Lambda_3)} \\ p(M_3|I_3, \Lambda_3) &= \frac{p(r_3|I_{3,2}, M_3)\lambda_3}{p(r_3|I_2, \Lambda_3)} \end{aligned}$$

- When breaks occur: trade-off between using longer history of data (more precise but biased) and shorter history (less precise but not biased)

back

Predictive Likelihood when λ is Estimated

For example: estimate of the predictive likelihood:

$$p(r_t|I_{t-1}) = \int p(r_t|I_{t-1},\lambda)p(\lambda|I_{t-1})d\lambda \approx \frac{1}{N}\sum_{i=1}^{N} p(r_t|I_{t-1},\lambda^{(i)})$$

Posterior moments with λ integrated out:

$$E[g(\theta)|I_t] = E_\lambda E[g(\theta)|I_t,\lambda] = \sum_{i=1}^{t} E[g(\theta)|I_{i,t},M_i]E_\lambda[p(M_i|I_t,\lambda)]$$

Note that $E_\lambda[\cdot]$ denotes expectation with respect to $p(\lambda|I_t)$

$$E_\lambda[p(M_i|I_t,\lambda)] \approx \frac{1}{N}\sum_{i=1}^{N} p(M_i|I_t,\lambda^{(i)})$$

back

News as a source of innovations to returns

Maheu and McCurdy (2004): Different types of news events have differential impacts on returns, on volatility & on higher-order moments of the return distribution

1. **Occasional significant news surprises (e.g. earnings surprises)**
   - Can induce large changes (jump innovations)
     - Intel, 12% increase on Jan.14, 2000 following positive earnings surprise
     - IBM, -17% on Oct.18,2000 following negative earnings surprise
     - Contributes to outliers (fat-tails) in returns distribution
   - But there is also evidence of jump clustering, for example
     - Intel, 10.7% on Apr.17, 2000, anticipating positive earnings surprise, then -8% on Apr.19, 2000 when positive earnings surprise was not as large as expected
     - Clustering of jumps also evident in the aftermath of the 1987 crash
     - Jump arrival clustering contributes to persistent conditional skewness & kurtosis
2. **Liquidity trading and strategic trading as news disseminates**
   - Smoother changes (normal innovations) contribute to GARCH or SV clustering
   - Previous jumps allowed to contribute to future volatility (increased uncertainty persists)

back

Parameterization of the Equity Premium: Background

Applying the 3-moment CAPM of Harvey-Siddique to the market as a whole

$$E[r_t|\Phi_{t-1}] = \eta_v E[\epsilon_t^2|\Phi_{t-1}] + \eta_s E[\epsilon_t^3|\Phi_{t-1}]$$

$E[\epsilon_t^2|\Phi_{t-1}] \equiv$ the conditional variance

$E[\epsilon_t^3|\Phi_{t-1}] \equiv$ the third central moment of market excess returns

back

Interpretation of our GARCH-jump mixture

Note that for our GARCH-jump mixture:

- Conditional on knowing # of jumps, the distribution is normal with two GARCH components plus the jump-size component
  - That is: $\sigma_t^2 + j\delta^2$
- λ_t is the mixing parameter: time-varying CPN distribution

In contrast:

- GARCH model is a mixture of Normal distributions over time
  - Affects unconditional distribution rather than conditional distribution. Given t-1 information, the conditional density is $N(\mu, \sigma_t^2)$
- Markov-Switching Models: mixture of discrete states
  - State is realized, take a draw
  - Transition probability matrix is the mixing variable (parameter)
  - Could rewrite our GARCH-jump mixture by defining infinite number of states so that it is a time-varying mixture of those states

back

Robustness Models: Specifications for Equity Premium

$$
\begin{aligned}
r_t &= m_t + \rho_1(r_{t-1} - m_{t-1}) + \rho_2(r_{t-2} - m_{t-2}) + \epsilon_{1,t} + \epsilon_{2,t} \\
\epsilon_{t-1} &= \epsilon_{1,t-1} + \epsilon_{2,t-1}
\end{aligned}
$$

	Pricing Models					
Parameter	GIM-1	GIM-2	constant λ	no $AR(2)$	prudence	intercept
μ	0.028 (4.146)	0.019 (2.785)	0.010 (0.731)	-0.051 (-2.255)		-0.015 (-0.866)
ρ_1	0.168 (22.515)	0.168 (22.216)	0.159 (22.180)		0.157 (21.909)	0.157 (21.879)
ρ_2	-0.029 (-3.889)	-0.032 (-4.350)	-0.048 (-6.864)		-0.048 (-6.958)	-0.049 (-7.000)
ψ_v	0.001 (0.111)	0.026 (2.711)	0.019 (1.678)	0.041 (3.380)	0.027 (3.270)	0.040 (3.266)
ψ_s			-0.014 (-1.029)	-0.112 (-2.923)	-0.028 (-3.438)	-0.054 (-2.395)
ψ_k			0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
lgl	-25512.470	-25305.875	-24627.911	-24825.400	-24577.290	-24576.271

back

Variance Dynamics

A 2-component GARCH specification augmented with asymmetric effects of jumps and bad news

$$
\begin{aligned}
\sigma_t^2 &= \sigma_{1,t}^2 + \sigma_{2,t}^2, \\
\sigma_{1,t}^2 &= \omega + g_1(A_1, \Phi_{t-1})\epsilon_{t-1}^2 + \beta_1 \sigma_{1,t-1}^2 \\
\sigma_{2,t}^2 &= g_2(A_2, \Phi_{t-1})\epsilon_{t-1}^2 + \beta_2 \sigma_{2,t-1}^2 \\
g_i(A_i, \Phi_{t-1}) &= \exp(\alpha_i + I(\epsilon_{t-1})(\alpha_{a,j,i} E[n_{t-1}|\Phi_{t-1}] + \alpha_{a,i})), i = 1, 2 \\
I(\epsilon_{t-1}) &= 1 \text{ if } \epsilon_{t-1} < 0, \text{otherwise} \quad I(\epsilon_{t-1}) = 0
\end{aligned}
$$

	Pricing Models					
Parameter	prudence	intercept	unrestricted	variance	skewness	kurtosis
ω	0.000	0.000	0.000	0.000	0.000	0.000
	(-0.700)	(-0.308)	(0.447)	(-0.383)	(-0.296)	(-0.585)
α_1	-4.798	-4.829	-4.874	-4.780	-4.827	-4.798
	(-26.895)	(-26.577)	(-26.840)	(-26.905)	(-26.620)	(-26.878)
β_1	0.974	0.976	0.978	0.973	0.976	0.974
	(357.003)	(333.603)	(380.423)	(355.490)	(333.507)	(352.002)
α_2	-8.254	-8.197	-8.139	-8.296	-8.213	-8.253
	(-3.515)	(-3.918)	(-3.616)	(-3.429)	(-3.684)	(-3.426)
β_2	0.760	0.767	0.780	0.753	0.767	0.759
	(35.475)	(38.291)	(48.587)	(32.052)	(38.363)	(34.476)
$\alpha_{a,j,1}$	-2.131	-2.357	-3.220	-1.935	-2.368	-2.077
	(-5.956)	(-5.000)	(-4.339)	(-6.601)	(-4.965)	(-5.882)
$\alpha_{a,1}$	1.603	1.519	1.373	1.656	1.514	1.613
	(8.083)	(6.693)	(6.818)	(9.124)	(6.657)	(8.136)
$\alpha_{a,j,2}$	-0.050	-0.119	-0.263	0.002	-0.121	-0.042
	(-0.511)	(-1.028)	(-2.617)	(0.020)	(-1.044)	(-0.428)
$\alpha_{a,2}$	6.222	6.266	6.369	6.179	6.284	6.207
	(2.648)	(2.990)	(2.830)	(2.552)	(2.815)	(2.573)

back

Robustness Models: Variance Dynamics

$$
\begin{aligned}
\sigma_t^2 &= \sigma_{1,t}^2 + \sigma_{2,t}^2, \\
\sigma_{1,t}^2 &= \omega + g_1(A_1, \Phi_{t-1})\epsilon_{t-1}^2 + \beta_1 \sigma_{1,t-1}^2 \\
\sigma_{2,t}^2 &= g_2(A_2, \Phi_{t-1})\epsilon_{t-1}^2 + \beta_2 \sigma_{2,t-1}^2 \\
g_i(A_i, \Phi_{t-1}) &= \exp(\alpha_i + I(\epsilon_{t-1})(\alpha_{a,j,i} E[n_{t-1}|\Phi_{t-1}] + \alpha_{a,i})), i = 1, 2 \\
I(\epsilon_{t-1}) &= 1 \text{ if } \epsilon_{t-1} < 0, \text{otherwise} \quad I(\epsilon_{t-1}) = 0
\end{aligned}
$$

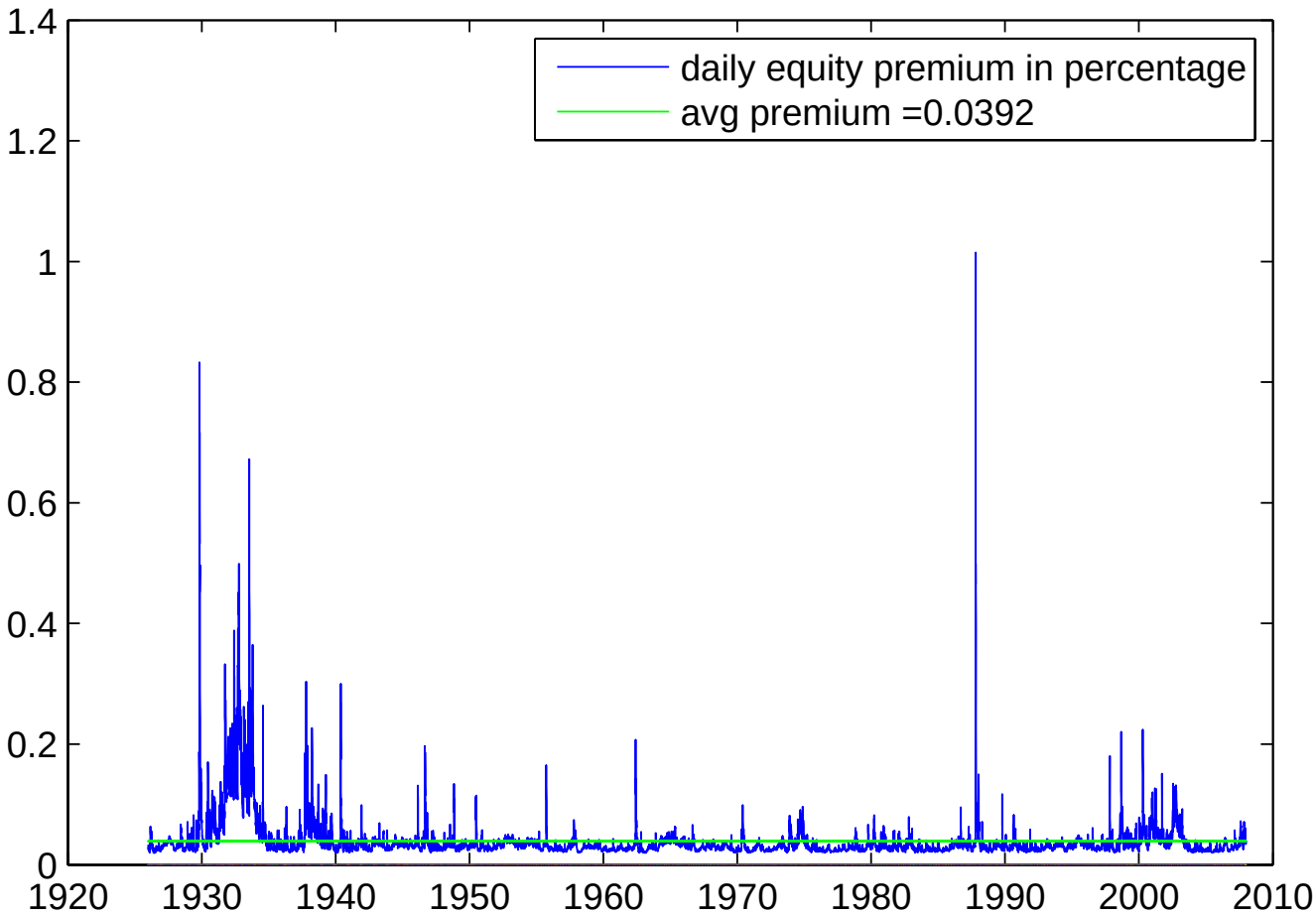
Parameter	GIM-1	GIM-2	constant λ	no $AR(2)$
ω	0.013 (12.524)	0.001 (6.512)	0.000 (1.614)	0.000 (-0.825)
α_1	-3.366 (-33.124)	-4.526 (-38.321)	-4.393 (-32.291)	-4.862 (-23.191)
β_1	0.895 (181.624)	0.984 (595.498)	0.974 (276.056)	0.978 (308.368)
α_2		-8.210 (-1.622)	-8.287 (-2.956)	-8.198 (-3.727)
β_2		0.798 (74.803)	0.793 (62.991)	0.776 (41.041)
$\alpha_{a,j,1}$			-6.439 (-1.568)	-2.601 (-5.959)
$\alpha_{a,1}$	1.432 (14.832)	-1.654 (-3.755)	0.691 (0.897)	1.395 (6.909)
$\alpha_{a,j,2}$			-0.588 (-6.310)	-0.194 (-1.745)
$\alpha_{a,2}$		6.604 (1.305)	6.834 (2.438)	6.302 (2.864)

Robustness Models: Jump Dynamics and Jump Size

$$\begin{aligned} \lambda_t &= E[n_t|\Phi_{t-1}], \quad \lambda_t = \gamma_0 + \gamma_1 \lambda_{t-1} + \gamma_2 \zeta_{t-1}, \\ \epsilon_{2,t} &= \sum_{k=1}^{n_t} Y_{t,k} - \theta\lambda_t, \quad Y_{t,k} \sim N(\theta, \delta^2) \end{aligned}$$

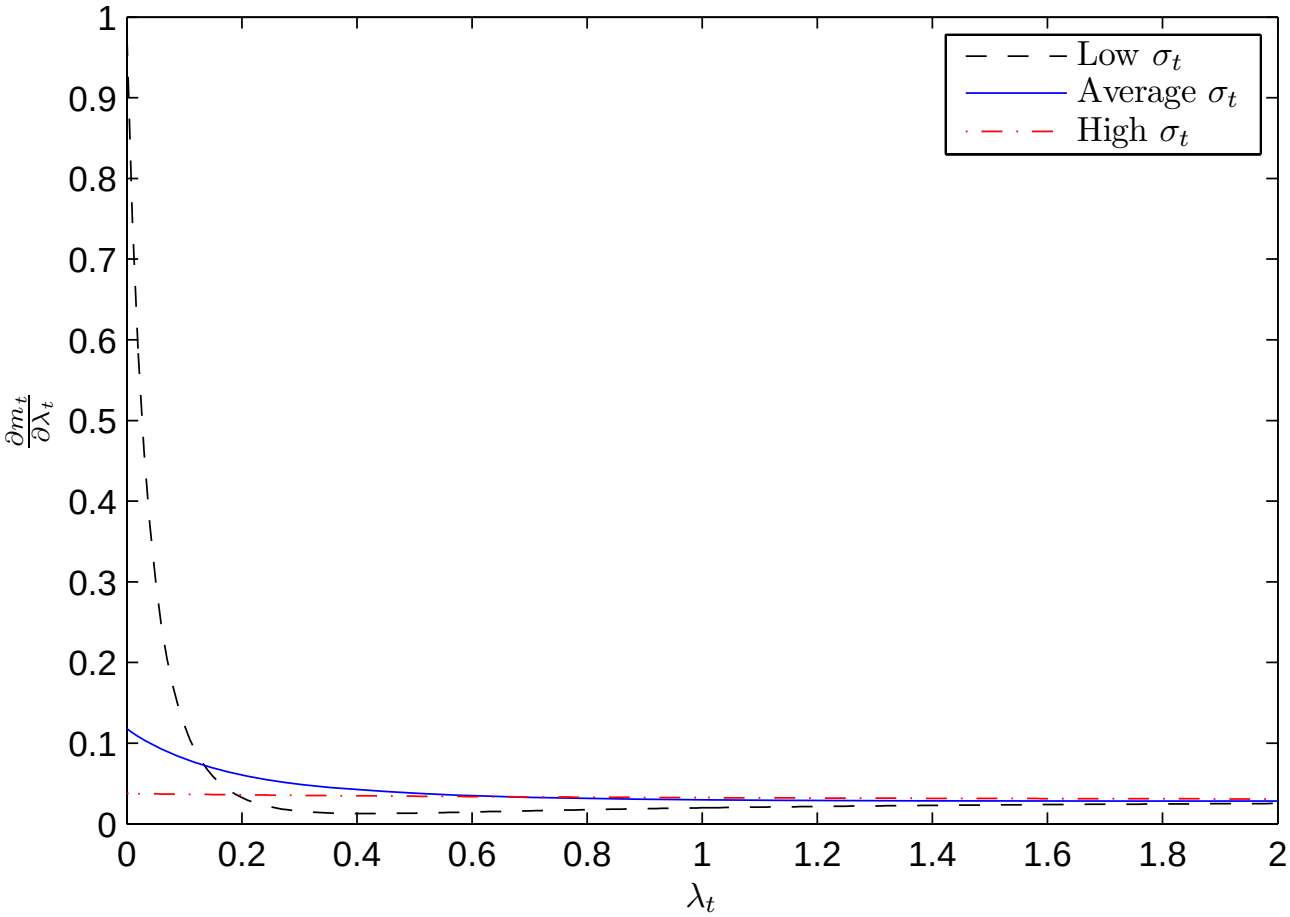
Parameter	GIM-1	GIM-2	constant λ	no $AR(2)$
γ_0			0.117 (5.858)	0.003 (2.195)
γ_1				0.974 (105.193)
γ_2				0.121 (4.348)
θ			-0.380 (-7.593)	-0.435 (-8.655)
δ			0.919 (14.265)	1.031 (15.330)

Daily Equity Premium Plot



back

Effect of Jumps on the Equity Premium



back

Summary Statistics for GIM-2 and Prudence Models

GIM-2 with intercept: Daily Equity Premium m_t and Variance v_t

	Median	Mean	StDev
m_t	0.032	0.046	0.053
v_t	0.496	1.029	2.038

GIM-2 without intercept: Daily Equity Premium m_t and Variance v_t

	Median	Mean	StDev
m_t	0.021	0.044	0.091
v_t	0.495	1.038	2.128

Prudence with Intercept: Daily Equity Premium m_t and Variance v_t

	Median	Mean	StDev
m_t	0.027	0.039	0.052
v_t	0.518	0.952	1.477

Prudence without Intercept: Daily Equity Premium m_t and Variance v_t

	Median	Mean	StDev
m_t	0.031	0.039	0.036
v_t	0.518	0.947	1.465